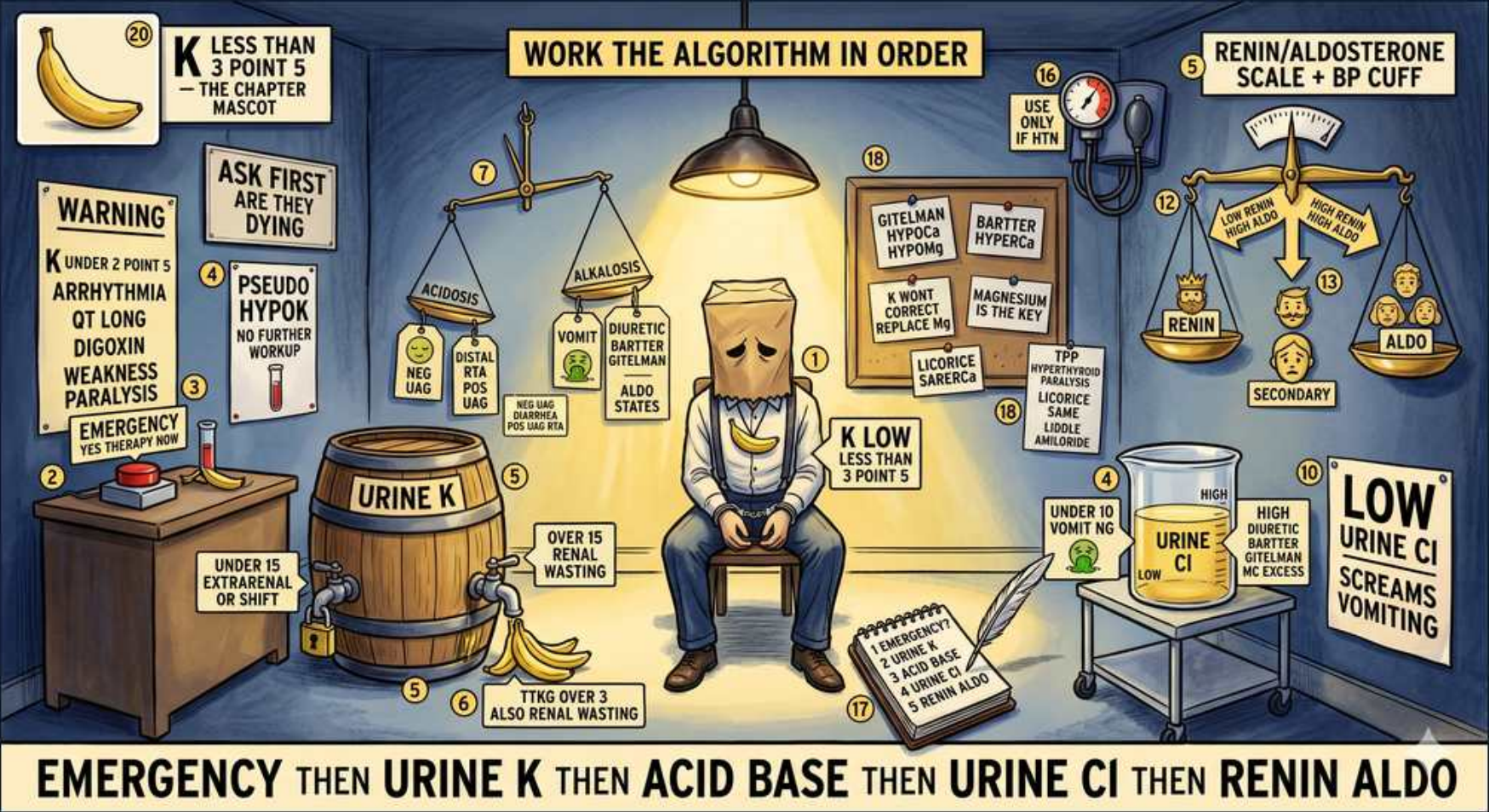


Hypokalemia — diagnosis



1. Work the order

WHAT YOU SEE

Banner: WORK THE ALGORITHM IN ORDER; bottom: emergency → urine K → acid base → urine Cl → renin aldo

WHAT IT ENCODES

Fixed diagnostic sequence for hypokalemia: (1) treat emergencies, (2) localize loss with urine K, (3) interpret acid-base status, (4) use urine Cl in metabolic alkalosis, (5) measure renin/aldosterone — but ONLY if hypertensive. Working in this order avoids ordering expensive hormonal tests before the cheap, decisive ones (urine K, BP, acid-base).

BOARD TRAP

WRONG: jumping straight to a renin/aldosterone ratio in any hypokalemic patient. Discriminator — renin/aldo is the LAST step and only relevant with HTN; localize first with urine K and acid-base. Skipping ahead is the classic mis-step.

2. Ask first: are they dying?

WHAT YOU SEE

Sign: ASK FIRST are they dying; emergency yes therapy now

WHAT IT ENCODES

Step 0 is triage: if there is arrhythmia, paralysis, or the patient is on digoxin, treat immediately — give potassium and place on a monitor — before completing the etiologic workup. Stabilization precedes diagnosis, mirroring all electrolyte algorithms.

BOARD TRAP

WRONG: delaying potassium repletion to finish the urine studies in an unstable patient. Discriminator — life-threatening features mandate treatment NOW; the cause can be sorted out after the patient is safe.

3. Emergency features

WHAT YOU SEE

WARNING K under 2.5: arrhythmia, QT long, digoxin, weakness, paralysis; emergency therapy now

WHAT IT ENCODES

Red flags forcing immediate IV K (with cardiac monitoring): K <2.5, arrhythmia or long QT, digoxin use, marked weakness or paralysis. IV potassium is given via a controlled rate (typically ≤10 mEq/h peripherally, ≤20 mEq/h central with monitoring) and Mg is replaced concurrently.

BOARD TRAP

WRONG: pushing rapid IV potassium boluses to correct severe hypokalemia fast. Discriminator — rapid IV K can itself cause fatal hyperkalemia/arrhythmia; rates are capped (≤10–20 mEq/h) with continuous ECG, never bolused.

4. Pseudohypokalemia

WHAT YOU SEE

Pseudo hypoK — no further workup

WHAT IT ENCODES

Before chasing a cause, exclude pseudohypokalemia: in marked leukocytosis (e.g. AML, CLL with very high WBC), metabolically active white cells take up potassium from the serum while the sample sits at room temperature, giving a falsely low value. Rapid processing or chilling the sample confirms it.

BOARD TRAP

WRONG: aggressively treating the low K reported in a leukemia patient with a sky-high WBC. Discriminator — suspect pseudohypokalemia in extreme leukocytosis; treating an in-vitro artifact risks real hyperkalemia. Recheck on a promptly processed sample.

5. Step: urine K

WHAT YOU SEE

URINE K barrel; under 15 = extrarenal or shift; over 15 = renal wasting

WHAT IT ENCODES

After emergencies, the FIRST localizing test is urine potassium: <15 mmol/day (or spot <15 mmol/L) means the kidney is appropriately conserving → extrarenal loss (GI/diarrhea) or transcellular shift; >15 means inappropriate renal wasting. This single test splits the differential before acid-base.

BOARD TRAP

WRONG: interpreting acid-base before checking where the K is being lost. Discriminator — urine K localizes renal vs extrarenal loss and should precede acid-base interpretation in the algorithm.

6. TTKG >3 = renal wasting

WHAT YOU SEE

TTKG over 3 also renal wasting

WHAT IT ENCODES

The transtubular potassium gradient (TTKG = [urine K × serum osm] / [serum K × urine osm]) or the simpler urine K:creatinine ratio refines the call: in hypokalemia a TTKG >3 (or K:Cr >13 mmol/g) signals inappropriate renal potassium secretion = renal wasting, when the kidney should be conserving.

BOARD TRAP

WRONG: interpreting TTKG when urine osm < serum osm or urine Na is very low. Discriminator — TTKG is only valid when urine is concentrated (Uosm > Sosm) and adequate distal Na delivery exists; otherwise it's uninterpretable. Many now favor urine K:Cr.

7. Acid-base — acidosis

WHAT YOU SEE

Acidosis side: distal RTA, pos UAG; vomiting nearby

WHAT IT ENCODES

Hypokalemia WITH metabolic acidosis narrows to: distal (type 1) RTA, proximal (type 2) RTA, or diarrhea/GI bicarbonate loss. Separate renal RTA from diarrhea using the urine anion gap (UAG = UNa + UK – UCl): RTA fails to excrete NH4Cl so UAG is positive; diarrhea has intact NH4 excretion so UAG is negative.

BOARD TRAP

WRONG: assuming all hypokalemia comes with alkalosis. Discriminator — hypoK + ACIDOSIS = RTA or diarrhea (use UAG: positive→RTA, negative→diarrhea); hypoK + alkalosis is a different branch entirely.

8. Acid-base — alkalosis

WHAT YOU SEE

Alkalosis bulb; vomiting, diuretic, Bartter, Gitelman, aldo states

WHAT IT ENCODES

Hypokalemia WITH metabolic alkalosis (the common pattern) splits by BP and urine Cl: normotensive → vomiting/NG suction, diuretics, or Bartter/Gitelman; hypertensive → mineralocorticoid excess (primary aldosteronism, Cushing, licorice, Liddle). Urine chloride and BP are the next discriminators.

BOARD TRAP

WRONG: lumping all hypoK + alkalosis as 'on a diuretic.' Discriminator — use BP (normal vs high) and urine Cl (low vs high) to separate vomiting/diuretics from Bartter/Gitelman from mineralocorticoid excess; don't stop at the alkalosis.

9. Magnesium — replace it

WHAT YOU SEE

Sign: K won't correct, replace Mg; magnesium is key

WHAT IT ENCODES

Always check and replace magnesium. Hypomagnesemia releases inhibition of the renal ROMK channel, causing ongoing renal potassium leak, so the K stays low no matter how much you give. Common with the same causes (diuretics, diarrhea, alcoholism, PPIs). Correcting Mg is mandatory in any refractory hypokalemia.

BOARD TRAP

WRONG: repeatedly redosing potassium in a patient whose K won't budge while ignoring Mg. Discriminator — refractory hypokalemia = check the magnesium; K cannot be corrected until coexisting hypomagnesemia is fixed.

10. Gitelman vs Bartter

WHAT YOU SEE

Gitelman hypoK hypoMg hypoCa; Bartter hyperCa

WHAT IT ENCODES

Inherited salt-wasting tubulopathies (both: hypoK + metabolic alkalosis + normal/low BP + high urine Cl, mimicking diuretics). Gitelman = thiazide-like defect (DCT NCC): hypomagnesemia + HYPOcalciuria, milder, presents in adults. Bartter = loop-like defect (TAL NKCC2): HYPERcalciuria ± nephrocalcinosis, presents earlier/more severe.

BOARD TRAP

WRONG: using serum calcium to tell them apart. Discriminator — it's URINE calcium that splits them: Gitelman = hypocalciuria (+ hypomagnesemia), Bartter = hypercalciuria. Think 'Gitelman = thiazide mimic, Bartter = loop mimic.'

11. Licorice / SAME

WHAT YOU SEE

Licorice / SAME box

WHAT IT ENCODES

Apparent mineralocorticoid excess: licorice (glycyrrhizic acid) and hereditary AME inhibit 11β-HSD2, so cortisol is not converted to cortisone and floods the mineralocorticoid receptor → HTN + hypokalemic alkalosis with LOW renin AND LOW aldosterone. Liddle (constitutive ENaC) gives the same low/low profile.

BOARD TRAP

WRONG: expecting high aldosterone in every hypertensive hypokalemic patient. Discriminator — AME/licorice/Liddle show LOW renin AND LOW aldosterone (cortisol or ENaC drives the receptor), versus primary aldosteronism (low renin, HIGH aldo). The aldo level distinguishes them.

12. Urine Cl step

WHAT YOU SEE

URINE Cl beaker; low urine Cl screams vomiting

WHAT IT ENCODES

In metabolic alkalosis, urine chloride (not urine Na) assesses volume status: LOW urine Cl (<20) = chloride-responsive — vomiting/NG suction or remote (off) diuretics; HIGH urine Cl (>20) = chloride-resistant — active diuretics, Bartter/Gitelman, or mineralocorticoid excess. This step directs saline-responsiveness.

BOARD TRAP

WRONG: using urine Na to judge volume in metabolic alkalosis. Why wrong — bicarbonaturia drags Na into the urine even when the patient is volume-depleted from vomiting, so urine Na looks falsely high. Discriminator — use urine Cl; low urine Cl 'screams vomiting.'

13. Renin/aldo + BP cuff

WHAT YOU SEE

Scale: renin/aldosterone, BP cuff, use only if HTN; secondary, low side

WHAT IT ENCODES

The final step, performed only if the patient is HYPERTENSIVE: the aldosterone-to-renin ratio names the mineralocorticoid disorder. High aldo + low renin = primary hyperaldosteronism (Conn); high both = secondary (renovascular, renin tumor); low both = AME/licorice/Liddle/Cushing. Interpreted alongside the high urine Cl from the prior step.

BOARD TRAP

WRONG: ordering renin/aldosterone in a NORMotensive hypokalemic patient. Discriminator — these tests are meaningless without established HTN; in normotensive hypoK the cause is GI loss, diuretics, shift, or a tubulopathy, not a primary mineralocorticoid disorder.